

Typed Falsifiability: Five Standing Predictions of the Trisductive Physical Seed, with Named Conditions and Bounded Consequence

A method for measuring how far a structural framework is empirically exposed, and its application to seventy-seven candidate predictions

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Whether a broad theoretical framework is empirically exposed cannot be settled by counting its predictions. A framework may publish sixty and remain untouched by all sixty failing, because none was load-bearing; may publish results already established in the literature, whose confirmation confirms the source and not the restatement; and may publish claims whose denial its own structure permits equally well, which are preferences rather than constraints. This paper develops a method that measures exposure directly. Each candidate prediction is passed through a six-step admission cascade and typed by four properties: whether the framework's structure forbids the denial, which named premises the prohibition rests on, what fails inside the framework if the prediction fails, and how many independent parties must agree before a falsifying result is established. The method is applied to the Trisductive Physical Seed, a substrate-monist arrangement of seven domain-level structures on a single actuation axiom. Seventy-seven candidates yield five standing predictions. Three further entries are structural necessities carrying no framework consequence; two are necessities belonging to general relativity and the Kerr geometry. The remaining sixty-seven are relocated with dispositions recorded and nothing discarded. No result carries a consequence larger than the retirement of one domain premise, though this is partly a property of how the register was assembled, since every candidate is seated within a domain; one failure branch carries less: a monopole attributable to substrate topology rather than to a failure of closure retires no domain at all and revises a substrate property whose reach is left open. No result reaches the framework's root, which is unreachable by a limitative theorem rather than by construction. Two of the five are settled by a single confirmed event in any adequately equipped laboratory, with no consortium in the falsification path.

KEYWORDS falsifiability, structural prediction, blast radius, topological constraint, positive kill conditions, warrant tiering, dark energy equation of state, monopole prohibition, baryon number conservation

Exposure is not established by enumeration

A structural framework organizes phenomena by geometric constraint rather than by added degrees of freedom, and the question of how far such a framework is empirically exposed does not have an obvious answer. The usual answer, a list of predictions, measures the wrong thing. It establishes that predictions can be written. It does not establish that any of them can break anything.

Three properties, each independently sufficient to hollow out a prediction list, are routinely left unmeasured. The first is

consequence. An entry that fails without taking anything with it has tested nothing, and a list may consist entirely of such entries while appearing substantial. The second is provenance. An entry drawn from established literature and restated in a framework's vocabulary carries the epistemic status of its source, so its confirmation confirms the source, and the restatement acquires no warrant by translation. The third is directional standing. An entry whose denial the framework's own structure permits equally well is a preference, however precisely its band is stated, because no result can contradict a structure that admits both directions.

Exposure is therefore a joint property of a prediction and the dependency structure above it. It cannot be established by care lavished on the prediction alone, and it can be measured only by examining what stands above each entry and what falls if the entry falls. This paper develops that measurement, applies it to a specific framework, and publishes the resulting distribution together with the surviving predictions.

The framework examined is the Trisductive Physical Seed: a substrate-monomist arrangement in which seven domain-level structures, here called residences, are seated on a single root axiom stating that existence carries substrate-level energetic content. Gravity, electromagnetism, the strong and weak interactions, elementary particles, the dark sector, and the black-hole horizon each enter as a residence with its own premises and its own boundary of validity. The arrangement, its axioms, and its open questions are set out elsewhere; ²¹ what is at issue here is which of its consequences can be broken, by whom, and at what cost.

Two questions that require different answers

A demand for falsifiability directed at a framework rather than at a claim conflates two questions. They have different answers and separating them is prerequisite to any measurement.

The root

The first question asks whether a framework's foundational commitment is itself empirically falsifiable. For this framework the commitment is a root axiom stating that to exist is to actuate, which in physical register means that existence carries substrate-level energetic content. Two distinct claims are often run together here and they are separated.

The first is that the root cannot be derived from a weaker base. That follows from Gödel's second incompleteness theorem and Tarski's undefinability theorem taken together: no sufficiently strong consistent system establishes its own grounding from within, and the unprovability is constitutive of foundation-hood rather than a property of any particular foundation.^{1,2} The theorem establishes underivability and nothing beyond it.

The second is that no observation bears on the root. Underivability does not license that, and the two relations are different: a claim can be unprovable from below and still be embarrassed by a measurement. The narrower and defensible statement is this. The universal extension of the energy floor over all existents whatever does no predictive work anywhere in this register. Every downstream entry rests on the confined-system bound, which is theorem-grade external physics rather than a posit. The root therefore carries no empirical exposure here because nothing here rests on its universal leg, and not because any theorem confers immunity on it. What the root would answer to elsewhere is outside this paper's scope and is not claimed either way.

The framework's checkable claims about its root are narrower. The energy floor is theorem-grade external physics for confined quantum systems, resting on the Heisenberg kinetic-energy bound and the zero-

point energy that follows from the canonical commutation relation, with the rigorous third law barring a cooled-to-zero state.³ The cost of any transition that occurs is charged by the Jarzynski and Crooks work relations, with Landauer erasure as the irreversible subclass, experimentally confirmed at the single-bit level.^{4,5,6} The universal extension of that floor over all existents whatever is a premise and is carried as one.

The consequence for the measurement is that the root is not a level in the consequence scale, because no entry in this register depends on it. Everything that predicts sits downstream on the confined-system bound, and everything downstream is exposed.

The residences

The second question asks whether a framework generates content that is empirically falsifiable, with consequence stated. Here falsifiability is graded and distributed. A framework is a nesting: a root grounding form and never content, an arrangement of residences on that root, and inside each residence a set of claims of differing provenance and structural standing. An empirical result strikes at some depth in that nesting, and the depth is a fact about the claim's dependency structure. Measuring and publishing that depth per entry is the method developed below.

Why forward predictions cannot be structurally sealed

One geometric result constrains what any such method may claim, and it is stated first because it bounds everything after it.

Consider a prediction as an object with three axes. Two are manifest and available now: the constraint the framework imposes at the future coordinate, and the trajectory of the present state propagated forward by identified dynamics. The third axis is the registration, and it is empty, because the measurement has not been made. Does the geometry of the two manifest axes constrain what will occupy the third?

It does not. Take the manifest axes as rows over N reading contexts, each centred to zero mean and then normalised to unit length in that order, with θ the angle between them. The identity below is insensitive to the order, holding to 1.3×10^{-15} under either convention, but the order is stated so that a check is unambiguous. Any unit witness W on the third axis decomposes into its projection onto the manifest plane and its orthogonal residual, with ρ^2 the squared norm of the residual. The Gram determinant of the completed triad is exactly

$$\det(R) = \sin^2(\theta) \cdot \rho^2,$$

so $\det(R) \leq \sin^2(\theta)$ for every unit witness, with equality precisely when the witness is orthogonal to the manifest plane. At θ equal to zero the manifest rows are dependent and both sides vanish, so the identity holds and carries no information; the method requires θ bounded away from zero and from π . For any two unit witnesses both orthogonal to that plane, ρ^2 equals one for each, so the determinant returns $\sin^2(\theta)$ for both, identically. **The determinant is constant on the orthogonal complement and carries no information**

distinguishing a genuinely sourced measurement from any other orthogonal direction.

Direct verification: on a fixed manifest plane at $\sin^2(\theta)$ equal to 0.859132, a witness aligned with an independent generator returned $\det(R)$ equal to 0.853265, while six witnesses drawn from noise returned 0.849903, 0.601481, 0.764135, 0.764666, 0.843447, and 0.760295. The sourced witness and the first noise witness differ by 0.0034.

The result holds where both manifest axes are populated from present data. Where a second axis is itself projected rather than measured, the manifest plane and θ are undefined and the inequality does not extend; such a claim carries two unmeasured axes and is outside the scope of this method.

Two consequences follow. No prediction may be sealed in advance by geometric argument, because the geometry that would do the sealing is blind to provenance; every entry below is therefore a permission plus a kill condition, never a lock. And the same blindness holds symmetrically: a dataset engineered to lie orthogonal to the manifest axes produces the identical determinant, so geometric agreement offers no protection against contaminated confirmation, which is why confirmation and falsification are accepted at different bars.

Why an orthogonality floor admits everything

A natural admission rule counts an entry as structurally supported when its two manifest axes lock. That rule fails on arithmetic.

Over two hundred thousand random centred vector pairs in twenty-four dimensions, $\sin^2(\theta)$ had mean 0.956 with a Monte Carlo standard error of 1.3×10^{-4} , median 0.979, first percentile 0.732, and minimum 0.367. The fraction of noise pairs clearing a threshold of one half was 0.9999 with binomial standard error 2.3×10^{-5} ; clearing eight tenths, 0.971 with standard error 3.7×10^{-4} ; clearing nine tenths, 0.866 with standard error 7.6×10^{-4} . Figures are reported at the precision the sampling supports. Two arbitrary directions in a space of modest dimension are near-orthogonal by construction, and an admission floor set there would admit every claim that can be phrased.

The discriminating test is therefore not whether the structure permits the claim but whether it forbids the denial. That is the test which reduces seventy-seven candidates to five.

Existing strategies and their structural limits

Five strategies are in use for establishing that a broad framework is falsifiable, and each fails on structure rather than execution.

Prediction volume. Answering by quantity fails because volume is orthogonal to exposure: a hundred consequence-free entries constrain a framework exactly as much as none.

Deferred scale. Locating the decisive test at an energy or length scale no foreseeable instrument reaches yields formal falsifiability with practical immunity. This has been the standing criticism of certain unification programmes.⁷

Parameter absorption. Expanding the parameter space to absorb each null preserves the framework at the cost of the property being claimed. The dark-matter direct-detection programme illustrates the pattern at field scale: three decades of nulls accompanied by repeated redrawing of the allowed mass-coupling region rather than closure.⁸

Band widening. Stating predictions at bands wide enough that survival is near-certain converts falsifiability into a formality. Widening is legitimate only when licensed by weakness of derivation, and the operational test of whether it was so licensed is whether any surviving entry is under adverse empirical pressure with its threshold held fixed.

Consequence silence. Stating predictions with full precision while omitting what depends on them leaves a reader unable to distinguish a load-bearing entry from a decorative one, and the distribution of consequence is exactly the quantity at issue.

Method

Three derivation conditions

A candidate enters the audit only if it satisfies three conditions drawn from disjoint instrument sets, so that agreement among them is informative rather than circular.

The first is formal-structural: a topological, geometric, or algebraic constraint must prohibit continuous variation or force a discrete threshold, and the constraint must be stated in terms invariant under the transformation group of the register in which the claim is indexed. A quantity that is an artefact of a chart selection rather than a property of the structure carries zero mutual information with the structure it is read off, and admitting such a quantity as a structural fact is a common failure in geometric argument. One candidate terminated on exactly this ground.

The second is thermodynamic-material: the constraint must produce a measurable signature at its boundary, and the signature must be a phase transition, a plateau, a power-law exponent shift, or a topological invariant rather than a scalar magnitude. A missing entity characteristically produces a bump, a recoil, or a resonance at a mass; a structural boundary characteristically produces a transition, a plateau, or an exponent that shifts once at a threshold and is invariant thereafter. The discriminant is typical rather than exceptionless, and it fails in both directions: production thresholds yield edge behaviour rather than resonances, and a first-order structural transition yields a latent-heat peak that reads as a bump. Where the discriminant is ambiguous the entry is typed on the negation test alone.

The third is empirical-observable: the test must be executable with instrumentation independent of the framework, using measurement principles orthogonal to the argument under test.

These three conditions are admission hygiene and should not be credited with the reduction reported below. Every one of the five surviving entries is a presence-or-absence claim rather than a plateau or an exponent shift, so the thermodynamic-material condition selected none of them. The filtering was done by the negation test at

the last step of the cascade, and the honest attribution is to that step alone.

All seventy-seven candidates cleared the entry gate, because the candidate set was assembled before these conditions were formalised and the conditions were applied downstream rather than at intake. The same holds of the cascade itself: its step order and its decisive negation test were formalised after the candidates existed and were not pre-registered against them. Neither the gate nor the cascade was blind to the material it filtered, and a reader should discount the reduction accordingly. What is pre-registered, and what a reader can hold the register to, is the kill conditions, which are fixed at publication and not revised against incoming data.

Falsification-route tiering

No falsification route may pass exclusively through a single proprietary dataset or a single coupled model. This requirement is sharpened into a tier travelling with each entry, since route strength is a property of the entry and not of the register.

A route is **E1** when a single confirmed event in any adequately equipped laboratory settles it, with no consortium, no combined posterior, and no shared reduction pipeline in the path. A route is **E2** when it requires a specific instrument class but multiple independent groups can and do run it with distinct systematics. A route is **E3** when it runs through a combined analysis or a consensus reduction, so that the result is mediated by a single methodological culture however many institutions sign it.

The tier states how many independent parties must agree before a result is established. An entry whose kill runs through one pipeline cannot carry terminal weight regardless of what that pipeline reports, and the tier is printed so that a reader can weight the entry accordingly.

Asymmetric acceptance

A falsification of a standing prediction is accepted at the lowest credible bar: a single confirmed positive result meeting the stated kill condition, reproduced once, retires the entry. No appeal to systematics and no reanalysis by the author.

The kill thresholds differ across entries because the observables differ in kind, and the difference is stated rather than left to be inferred. Predictions II through V are discovery claims, a positive detection of an object or a mode, and take the five-sigma convention of that literature. Prediction I is a parameter-estimation claim about a posterior boundary crossing and takes the three-sigma convention of the cosmology literature against which it is scored. The bars are not interchangeable and equalising them would misapply one convention to the other.

A confirmation is accepted only when the shared-source structure of the supporting evidence has been examined and independent modalities agree. Agreement among measurements sharing an upstream source is one voice in several instruments, and consensus within a theoretical culture is not independent evidence. The criterion is pre-commitment rather than intent, since intent is unobservable: a

measurement whose analysis plan and threshold were fixed before the data were unblinded counts as a test, and one whose plan was selected after inspection does not, whichever way its result falls. Weight therefore accumulates on entries that cut against current expectation rather than with it.

One clause bounds the rule. A contamination objection is admissible only on an exhibited, independently reproducible methodological defect. Disagreement between a result and a prediction is never itself evidence of contamination.

Pre-commitment and retention

Every kill condition is fixed at publication with a date, deposited with the archived version, and not revised in response to incoming data. Where a result lands between a stated confirmation band and a stated kill band, the entry is reported as neither and the gap is reported as a defect in the original statement.

No candidate is discarded at any stage. Entries that fail are relocated with disposition, class, consequence level, and route tier recorded in a ledger published alongside, so that the full distribution and the reasons for each termination remain available for inspection.

The admission cascade

The procedure is mechanical given its inputs, the inputs are declared, and the emitter fails safe: an under-specified candidate cannot reach the strongest class.

Step one, provenance. Native to the framework, or drawn from literature and restated? An imported claim carries its source's status. The refinement that matters is whether the necessity is independently re-derivable without the source; if it is, the literature is corroboration and the candidate proceeds.

Step two, term and frame audit. The claim is decomposed into existence, kinetic, and relational components, and the vocabulary populating the three must be pairwise disjoint. A term carrying two referents inside one statement halts here.

Step three, domain gates. Twelve directed consistency checks, of which one is decisive in practice: a result transported across a domain boundary without a stated and defensible bridge premise terminates.

Step four, invariance audit. The predicted quantity is checked against the invariant ring of its register's transformation group. A magnitude manufactured by a chart selection, a width, an address, a mid-position, is barred from entering as a structural fact even when it is measurable and even when it is correct.

Step five, admission. The claim must enclose a genuine multi-axis volume with the formal axis among the axes, and must be directed, since a squared magnitude is invariant under reflection of any axis and carries no direction. Magnitude without direction routes to permission.

Step six, negation. The decisive filter. If the framework's structure equally permits the denial, the entry is a lean rather than a necessity. If the denial is forbidden, the premises on which the prohibition rests

are counted and must be printable. Every necessity reported here is conditional in this exact sense, holding on premises that are enumerated with the entry; a necessity asserted with no premise named is barred as unconditional and does not appear.

Consequence levels

Each surviving entry states what fails inside the framework if it breaks, on a scale fixed by the dependency structure. The face and premise distinction is operational: delete the claim from the residence and ask whether the residence’s remaining claims are still derivable. If they are, the claim is a face and its failure is **L1**, the residence and the framework standing with a hole named. If they are not, the claim is a premise and its failure is **L2**, that residence retired and the remaining six plus the substrate reading standing. **L0** marks an entry on which nothing in the framework depends, which is a statement about dependents and not about derivation: an entry may be a strict structural necessity and still be L0 if no other claim rests on it. **L3** marks the arrangement of residences itself. No candidate in this register could have reached it, because every candidate is seated within a residence and none quantifies over the arrangement, so the absence of L3 outcomes is a property of how the register was built and is reported as a scope note rather than as a finding about the framework. The root is not a level, for the reason given above.

Results

Table 1 | Disposition of all seventy-seven candidates. Seventy-two identifiers entered; one merged, two split into four, four newly derived from residence premises, giving 72 – 1 – 2 + 4 + 4 = 77. Discarded: none.

Disposition	Count	Weight carried
Standing predictions	5	The framework’s exposure
Method exhibits, L0	3	Structural necessity, framework consequence nil
Inherited necessity	2	Necessity belongs to general relativity or the Kerr geometry
Relocated, structural lean	24	Denial not forbidden; no terminal weight
Relocated, corroboration	38	Carries the source’s status, not the framework’s
Terminated, out of band	4	Bridge unestablished or magnitude chart-manufactured
Repaired, then relocated	1	Halted at term audit; repaired and rehoused
Total	77	

Per-residence yield is uneven and is reported with its denominator, since a yield without the count of candidates entered cannot distinguish a residence that was probed and came up empty from one

that was barely probed at all. Gravity took eleven candidates and yields two standing entries, one of them shared with the horizon. Electromagnetism took four and yields one. The strong interaction took three and yields one. Elementary particles took six and yields one. The horizon took five and yields the one it shares with gravity. The weak interaction took two and yields none. The dark sector took nine and yields none of its own: its one surviving entry is seated at gravity, because the premise that fails under a confirmed crossing is the fluid reading of the tension rather than any dark-sector claim. The remaining thirty-seven candidates were seated at no residence at all, which is itself the register’s largest single fact about its own composition. Two of seven residences therefore contribute no exposure, and one of those two was among the more heavily probed.

Table 2 | The standing five. No entry reaches L3, which follows from every candidate being seated within a domain rather than from a property of the framework. Prediction II carries two failure branches at different consequence levels, marked in its row and set out in the entry.

Prediction	Residence	Level	Route	Status
Phantom divide never crossed	Gravity	L1	E3	Under active test, in tension, threshold unmet
No magnetic monopole	Electromagnetism	L2 or untyped	E1	Standing on absence
Purely quadrupolar radiation	Gravity and horizon	L2	E2	Constrained, not settled
Baryon number conserved	Strong	L1	E1	Standing on absence
No electron substructure	Particles	L2	E2	Standing on absence

Box 1 | Four terminations and the step at which each occurred

A biophysical efficiency claim halted at the term audit. The candidate set its kill condition at a transduction efficiency above ninety-seven percent, described as violating the classical thermodynamic bound. It does not; the bound on free-energy transduction at fixed chemical potentials is one hundred percent, and the word efficiency was carrying two distinct referents within a single statement. Repaired and relocated to corroboration.

A galactic acceleration-scale claim terminated at the invariance audit. The candidate held that the low-acceleration scale coevolves with the Hubble parameter as a geometric feature of the expansion metric. Under the framework's own typing that scale is a family rather than a member, because the substrate carries no temperature, no fluctuation spectrum, and no coupling with which to pin a diffusivity, and the numerical coincidence between that scale and the product of the speed of light with the Hubble constant is a chart-manufactured magnitude rather than an invariant. Terminated out of band. The empirical radial-acceleration regularity, a datum and not a derivation, is retained separately as corroboration.¹⁷

Three candidates terminated at the domain gates: the Weak Gravity Conjecture applied to relic abundance, the Bekenstein bound applied to quark-gluon plasma, and the ER=EPR correspondence applied to a laboratory teleportation protocol. Each requires a bridge premise the framework does not establish and cannot supply.

Twenty-four candidates failed the negation test. In each the structure locks and points, but the denial is not forbidden. They are informative if confirmed and carry no terminal weight, and are relocated as leans rather than presented as constraints.

The standing predictions

Each states the structural claim, the premises on which the prohibition rests, the method, the positive kill condition, the consequence level, the route tier, and the current status. Four currently stand on absence, which confirms nothing and bounds the effect below the sensitivity reached; each says so.

Prediction I · The phantom divide is not crossed

Structural claim. Reading dark energy as the residual tensional gradient of a continuous substrate rather than as a vacuum constant forces the equation-of-state parameter to satisfy $w(z) \geq -1$ at all redshifts.

Conditions, two. That the tension behaves as a real fluid, and that the null energy condition holds for that fluid. If either fails the prohibition lapses, and the failure is informative about which.

Method. Combined posteriors from DESI baryon acoustic oscillations, Euclid weak lensing and BAO, LSST supernovae, and Pantheon+, under both a w_0 - w_a CDM parameterization and a model-independent reconstruction, with the crossing redshift cross-validated across independent supernova compilations.

Kill condition. A confirmed crossing of w equal to -1 at greater than three standard deviations in a shape-free reconstruction of $w(z)$, by which is meant a Gaussian-process reconstruction or a binned or nodal reconstruction with a correlation prior, rather than a fit to a monotone two-parameter form.

Consequence, L1. One face of the gravity residence. The substrate reading, the emergence of the field equations as a thermodynamic equation of state,¹⁰ and the remaining six residences stand. What fails is the fluid reading of the tension, or the null energy condition for it, and the register is obliged to report which.

Route tier, E3. The falsification runs through combined survey posteriors, so it is consensus-mediated and cannot carry terminal weight in either direction.

Box 2 | The status of Prediction I, and why the parameterization is the question

This is the only standing prediction currently under active test against data, and it is in tension.

Combined baryon-acoustic-oscillation, cosmic-microwave-background, and supernova analyses favour an evolving equation of state over a static cosmological constant, with the preferred fit sitting at roughly 2.6 to 3 standard deviations from the constant case.⁹ That figure is a preference for evolution over a constant and is not the crossing significance the kill condition names; the two are computed on different hypotheses and their numerical proximity is coincidental. Under the two-parameter Chevallier-Polarski-Linder form, $w(z) = w_0 + w_a z/(1+z)$, a fit returning w_0 above minus one together with w_a negative implies a crossing of the phantom divide by construction, since the function is then monotonically decreasing in redshift from a starting value above minus one. The crossing in that setting is a property of the fitting form rather than an independent measurement of it.

Model-independent reconstructions, which impose no shape, carry substantially larger uncertainty and do not establish the crossing at the threshold this prediction names. The prediction therefore stands, in tension, with the decisive measurement identified.

The threshold is not moved and no condition is added. If the crossing is confirmed at that bar the prediction breaks, one face of the gravity residence is retired, and the register reports which of its two premises failed.

Prediction II · No magnetic monopole

Structural claim. Electricity and magnetism are two frame-borne projections of a single two-form on the substrate. A Lorentz boost rotates one face into the other, so they are one object and not two forces. This is standard electrodynamics and discriminates nothing on its own; it is used here only to fix that the two faces occupy one domain rather than two, and the predictive content of the entry sits entirely in the two premises stated below it. Closure of the two-form kills the co-exact sector, leaving an exact part carrying standard electromagnetism and a harmonic part carrying topological flux. Closure alone does not forbid isolated magnetic charge: it removes the

magnetic source term, but a harmonic two-form with non-zero period through a closed surface is precisely a magnetic flux, and such forms exist whenever the second cohomology of the substrate is non-trivial. The prohibition therefore requires that the harmonic sector carry no net flux through any closed surface, equivalently that the substrate's second cohomology be trivial, so that every closed two-form on it is exact.

Conditions, two. That the substrate two-form is closed, and that its second cohomology is trivial. A confirmed monopole retires one of the two, and the register is obliged to report which, since the two failures carry different consequences: a failure of closure retires the residence, while a non-trivial second cohomology leaves the two-form reading intact and relocates the charge to the substrate's topology.

Method. Any monopole search whatever, including superconducting induction detectors, accelerator searches, cosmic-ray detectors, and searches in bulk matter. No specific facility is named because none is required.

Kill condition. A single confirmed isolated magnetic charge, reproduced once by an independent group.

Consequence, L2 on one branch, untyped on the other. If closure fails, the electromagnetism residence is retired, the remaining six residences and the substrate reading stand with a hole named where electromagnetism was, and this is the largest consequence any single result here carries. If instead the second cohomology proves non-trivial, no residence is retired at all: the two-form reading survives and what changes is a global property of the substrate. That branch is not typed by the L0 to L3 scale, which is defined over residences, and whether any residence other than electromagnetism depends on trivial second cohomology is not established here in either direction. The reach of that branch is open, and the entry says so rather than assigning it a level it has not earned.

Route tier, E1. A single event in any adequately equipped laboratory settles it. No consortium, no combined posterior, no shared reduction pipeline stands in the path.

Current status. Standing on absence. No confirmed monopole has been observed, and that absence confirms nothing; it bounds the flux below the exposure reached. The entry is a prohibition rather than a rate prediction, so no flux sensitivity threshold attaches to it and none is stated: a single confirmed charge kills it at any flux, and no accumulation of null exposure ever confirms it.

Prediction III · Gravitational radiation is purely quadrupolar

Structural claim. The rank-2 source structure forces a spin-2 mediator by exhaustion. For a massless symmetric rank-2 field in four spacetime dimensions the component count closes at ten minus four gauge minus four constraint equal to two, and two polarization states, and only two, are permitted; the massive case, which carries five, is the subject of the classical Fierz-Pauli analysis.¹² Breathing modes and vector modes are forbidden. The count is dimension-dependent and mass-dependent: in D dimensions it returns $D(D+1)/2 - 2D$, giving

zero at D equal to three, two at four, five at five, and nine at six, and a massive spin-2 field in four dimensions carries five states rather than two.

Conditions, three. That the source structure is rank-2, that the mediator is massless, and that spacetime is four-dimensional at the scales probed. The first two are co-tested by the polarization measurement itself. The third is a background assumption rather than a co-tested premise, since a detector network cannot distinguish four dimensions from four plus compactified extra dimensions invisible at its scale, and a confirmed non-tensor polarization therefore adjudicates between the first two while leaving the third to be settled elsewhere.

Method. Polarization decomposition of the gravitational-wave signal with a detector network of sufficient geometric diversity to separate the six possible metric polarizations. The Einstein Telescope and Cosmic Explorer reach the required sensitivity and network geometry; current-generation networks constrain but do not settle it.¹⁴

Kill condition. Confirmed detection of a breathing mode or a vector polarization mode at greater than five standard deviations in a single well-localized event with adequate network coverage.

Consequence, L2 spanning two residences. The spin-2 exhaustion argument is load-bearing for the gravity residence, and the horizon residence inherits it for the quadrupolar radiation requirement. A confirmed scalar or vector polarization retires the exhaustion argument in both. The substrate reading and the remaining five residences stand.

Route tier, E2. A specific facility class is required, but multiple independent collaborations with distinct instruments and systematics can run the analysis, and the polarization decomposition is geometric rather than model-dependent.

Current status. Constrained, not settled. Present networks lack the geometric diversity to separate all six polarizations decisively, so current consistency with tensor-only content is a bound rather than a confirmation.

Prediction IV · Baryon number is exactly conserved

Structural claim. Perturbative baryon-number conservation is shared with the standard model and discriminates nothing on its own; what this entry adds is a prohibition at every reachable rate, which cuts against grand-unified extensions predicting decay within current detector exposure rather than against the standard model. A baryon is a closed three-strand braid configuration, and its baryon number is the configuration-level Hopf invariant. A topological invariant of a closed configuration is conserved by a classical theorem, and a change of sector requires a non-perturbative transition through a configuration that is not closed. No suppression exponent is quoted here, because the artifact carries no executed calculation of the relevant action and a recalled figure would not be checkable. The claim is therefore the qualitative one the topology supports: sector change is non-perturbative, and proton decay and neutron-antineutron oscillation are forbidden at perturbative rates.

Conditions, two. That the configuration is genuinely closed, and that the identification of baryon number with the Hopf invariant holds.

Method. Hyper-Kamiokande and DUNE for proton decay across all accessible channels; NNBAR for neutron-antineutron oscillation.

Kill condition. A confirmed proton decay event in any channel, or a confirmed neutron-antineutron oscillation, at discovery significance with independent reproduction.

Consequence, L1. The identification of baryon number with the Hopf invariant is retired within the strong-force residence. The geometric account of confinement, which rests on the cost of an open strand rather than on the invariant, stands independently, as does the residence with the identification removed.

Route tier, E1. A single confirmed decay event settles it. The detectors are large collaborations, but a discovery-significance event is not a combined-posterior inference.

Current status. Standing on absence. Existing partial-lifetime bounds constrain the rate and confirm nothing.¹³

Prediction V · No electron substructure below the stated scale

Structural claim. Elementary particles are knot configurations in a single substrate field, with the electron identified as the minimal non-trivial configuration. A minimal configuration has no constituents, so the electron carries no substructure and no form factor beyond the quantum-electrodynamic prediction.

Condition, none separable. This entry alone carries no premise distinct from the domain it belongs to, because the claim that particles are knot configurations is the domain rather than an assumption beneath it. The entry is therefore unconditional within its domain: a confirmed form factor or compositeness signal retires the domain outright, with no branch on which some other premise absorbs the failure. That is what its consequence level already records, and no condition is manufactured here to make the entry match the shape of the other four.

Method. Precision measurement of the electron anomalous magnetic moment tested for a form-factor deviation, which is executable now and is the near-term route; and direct compositeness searches at the highest accessible collider energies, whose present reach covers the lower part of the twenty to one hundred teraelectronvolt scale and whose upper part awaits a higher-energy machine.

Kill condition. A confirmed electron form factor at discovery significance, which the anomalous-moment route can deliver with present instruments; or a confirmed compositeness signal at discovery significance, reachable today only to the lower end of the stated scale, with the upper end horizon-limited by collider energy rather than by the prediction.

Consequence, L2. The particles-as-knots residence is retired, since the claim that every particle is a knot in the one field is the residence

rather than a face of it. The substrate reading and the remaining six residences stand.

Route tier, E2. A specific facility class is required and the result would be produced by a large collaboration, but the anomalous-moment route and the collider route are independent modalities with unrelated systematics.

Current status. Standing on absence. Existing compositeness limits bound the scale and confirm nothing.

Necessities carrying no framework weight

Three further entries reached necessity at consequence level L0: the tetrahedral defect configuration on spherical nematic shells, the robustness of topological surface states below the percolation threshold, and the quantized Majorana conductance at two e^2/h .^{15,16} All three are genuine necessities and all three carry an E1 route, but none carries any framework consequence, so none tests the framework.

Two entries are necessities belonging elsewhere: the gravitational-wave memory effect, a prediction of general relativity, and photon-ring substructure with sub-ring ratios fixed by the Kerr geometry. The framework inherits both, so their confirmation confirms the source, and they carry no framework weight.

Discussion

The shape of the distribution

Five of seventy-seven candidates survive as standing predictions, a yield near six percent. Of the seventy-two that do not, thirty-eight belong to the literature and are corroboration within a formalism, twenty-four are cases where the structure permits the claim but does not forbid its denial, four transport across a domain boundary without a bridge, one halted on a term defect, and five are necessities carrying no framework consequence or belonging to other theories.

The distribution is itself the principal empirical finding about the framework. A framework whose register is dominated by inherited results and undirected leans is exposed by very few of them, and identifying which few is the whole of the measurement.

The shape of the consequence

No standing entry reaches L3. The largest consequence any single confirmed result carries is the retirement of one residence, carried by three entries: the monopole, the polarization structure, and the electron compositeness. Two carry L1, retiring one face of one residence.

Two readings follow and both are available from the same table. A framework whose entire exposure fits on two pages is not heavily exposed. A framework whose exposure is precisely located, stated in advance with kill conditions a stranger can execute, and bounded so that a reader knows exactly what each result costs, is exposed in the only way that supports experimental engagement. The measurement

supports both readings and distributes the information required to hold either.

Standing on absence

Four of the five currently stand on absence, and absence at finite sensitivity is consistent both with the effect being absent and with the instrument failing to reach it. None of the four is confirmed by continued nulls and each states so. What each does is name a positive, reachable, currently unobserved result that breaks it, and two of those results can be produced by a single laboratory.

The fifth is under active test and in tension. The tension isolates a live methodological question, whether a crossing inferred from a monotone two-parameter form measures the crossing or the form, and it commits the entry to a threshold stated in advance and reachable by an analysis the community is already building.

The protected root, and its bound

The root is not in the consequence table because no empirical result places it there, and a protected root is a permanent asymmetry in a framework's favour. The asymmetry is real and is bounded rather than denied. The root grounds form and never content: it donates no physics, fixes no constants, selects no gauge group, and predicts nothing on its own. Everything that predicts is downstream and exposed. A protected root that donates nothing shields nothing.

The discriminating outcome

Survival is the weakest outcome and establishes little, since a sufficiently cautious register survives by construction. A discriminating outcome requires a standing prediction confirmed in a regime where a competing programme predicts something observably different. Prediction III is the clearest such case: a network-geometry polarization measurement returning strictly two modes, against alternative-gravity programmes predicting scalar or vector content at accessible amplitude, discriminates directly.

Limitations

The class assignments are hand-built inputs to a mechanical emitter. The emitter is deterministic and its defaults fail safe, but the inputs are readings, and a reader who types an entry differently obtains a different disposition. The emitter, every declared input, and the disposition ledger are deposited with the archived version of this paper and are listed under Data and reproduction, so that a reader may re-run the typing and contest any assignment.

One consequence branch is untyped. A monopole attributable to non-trivial second cohomology rather than to a failure of closure revises a substrate property instead of retiring a residence, and the consequence scale used throughout is defined over residences, so it does not reach that case. Whether the remaining six residences depend on trivial second cohomology is not established here, and the question is open rather than answered in the framework's favour.

Every numerical exhibit here is single-seed, single-script, and single-author. The exhibits are reproducible, and reproducibility is not

independence: no party other than the author has re-run them. The open question is not whether they reproduce but whether an independent execution agrees, and it is settled only by that execution.

The framework was synthesized from the published literature and cannot fully audit its own priors from within. Where a prediction agrees with current theoretical consensus, the apparent structural agreement may be one culture speaking three times. Cross-domain falsifiability raises the cost of a false convergence by requiring agreement across communities that share no instruments or referees, but it does not eliminate the risk.

The boundary between necessity and lean rests on whether the structure forbids the denial, and that judgement is not always sharp. Three relocated entries sit near the boundary and were assigned to lean on the conservative side.

The relocated sixty-seven are not discarded knowledge. Several are among the most immediately executable claims available, and they are worth running on their own merits. What they cannot do is carry the framework's weight.

The relocations themselves were typed by the cascade but were not adversarially prosecuted. The five standing predictions were; the sixty-seven were not, and their dispositions therefore carry the author's typing without an opposing pass. A relocated entry that in fact forbids its own denial would have been missed, and the direction of that error runs against the framework rather than for it, since it would understate the exposure. The ledger is published so that a reader may re-type any relocated entry and contest the disposition.

Conclusion

Whether a structural framework is empirically exposed is a measurable property, and the measurement requires typing each of its predictions by structural standing, provenance, consequence, and falsification route rather than counting them.

Applied to the Trisductive Physical Seed, seventy-seven candidates yield five standing predictions, three necessities carrying no framework consequence, two necessities belonging to general relativity and the Kerr geometry, and sixty-seven relocations with dispositions recorded and nothing discarded. The largest consequence any single result carries is the retirement of one residence premise. No result reaches the framework's root, for a reason that is a limitative theorem rather than a design choice.

Two of the five, the monopole prohibition and baryon-number conservation, are settled by a single confirmed event in any adequately equipped laboratory, with no consortium and no shared reduction pipeline in the falsification path. The experimental community is invited to execute them. Falsification is accepted at the lowest credible bar; confirmation is accepted only after an independence audit.

The most consequential open question is not any of the five. It is whether the substrate can be characterized, because until it is, the framework inherits every number from the physics it seeks to ground, and its honest predictions concern structure rather than magnitude.

What follows for the general practice is a modest reframing: falsifiability is a graded property distributed across a framework's dependency structure rather than a binary property of the framework as a whole, and a register that does not publish the distribution has not established the property, whatever its length.

Appendix A · Foundational axioms, stated as standalone principles

The following axioms are drawn from a broader epistemic framework and are presented as standalone physical principles, each independently motivated and independently testable within the native discipline.

The actuation floor. Existence carries substrate-level energetic content: for any confined system the kinetic energy is strictly positive, and any transition that occurs is charged a thermodynamic cost bounded below. The first clause is theorem-grade for confined quantum systems on the kinetic-energy bound and the zero-point energy; the second is the content of the fluctuation theorems, with Landauer erasure as the irreversible subclass. The extension of the floor over all existents whatever is a premise.

Substrate monism. Substrate, topology, and actuation are three projections of one event rather than three independent posits. The physical content is the continuous-field ontology with smeared field operators; the priority of the one over the many is a structural posit carried as a premise.

Register invariance. A geometric claim asserts only what is invariant under the transformation group of the register in which it is indexed. A magnitude produced by a chart selection carries zero mutual information with the structure it is read off and does not enter a verdict as a structural fact. This is the Erlangen criterion applied as an admission gate.

Orientation blindness. A squared magnitude is invariant under reflection of any axis, so it certifies the dimensionality of a structure and never its direction or truth sign. Direction is read from the directed content of the claim and from the thermodynamic arrow, never from the magnitude.

The completion inequality. For a claim with two established axes and one unmeasured axis, the joint determinant satisfies $\det(R) = \sin^2(\theta) \cdot \rho^2$ and is bounded above by $\sin^2(\theta)$, attaining that bound for every witness orthogonal to the established plane. The determinant is constant on the orthogonal complement and carries no information about the provenance of the third axis. The measurement occupies it.

Appendix B · Disposition ledger

Sixty-seven candidates did not reach standing and none was discarded. The ledger published alongside records for each its identifier, disposition, class, consequence level, route tier, and locator, so that any relocated claim can be reconstructed and contested.

Dispositions are five. **Structural lean**, twenty-four entries: the structure locks and points but does not forbid the denial, so the entry is informative if confirmed and carries no terminal weight.

Corroboration, thirty-eight entries: the claim belongs to the literature and the structural restatement is corroboration within a formalism, carrying the source's status. **Out of band**, four entries: one terminated at the invariance audit and three at the domain gates. **Repaired then relocated**, one entry halted at the term audit and rehoused after repair. **Merged**, one identifier folded into two surviving entries.

The census reconciles arithmetically: seventy-two identifiers entered; one was merged into two survivors, removing one card; two identifiers each carried two separable claims and were split, removing two source cards and adding four; four cards were newly derived from residence premises. Seventy-two minus one minus two plus four plus four is seventy-seven. None discarded.

Appendix C · Author's provenance and method disclosure

This paper was developed under Trisduction, a verification and organizing discipline, not a source of results. Its conclusions rest solely on the standard results cited in the body. Trisduction is the method under which those results were decomposed, assembled, and audited. It adds them no warrant and claims no authorship over them.

Trisduction was used for fidelity. It forces a claim onto three independent axes so no single persuasive line carries it alone, requires every verdict to resolve to one of three states, sealed, broken, or open, each with a named failure mechanism, and attaches an explicit warrant grade, theorem, conditional, structural, or premise, to every claim so nothing is stated above its strength. Two errors it is built to catch are inflation, reading an internal lock as a proof, and circularity, reading a restatement of a claim as a derivation of it.

The root axiom is that to exist is to actuate: every physical existent carries a strictly positive energetic cost, grounded in the Heisenberg energy floor, the zero-point energy, and Landauer's bound. The formal root inside it is that to formally be is to be grounded, with provability and computation levels of access to a determinacy fixed at the ground rather than ingredients of it.

The key procedures, in brief. Orthogonal triaxial convergence: a proposition is split into three disjoint axes, formal-structural, empirical or dynamical, and registrational, verified by three separate instrument sets, agreement across the three the operational meaning of a seal. The geometric orthogonal lock: the three axes are read as vectors and their independence is tested by the determinant of their correlation matrix, a determinant clear of zero a genuine three-dimensional lock, a collapse to zero one axis dissolving into the plane of the other two. The convergence dissolution test: before the lock is read, any mass-bearing common cause the three axes might share is projected out, so an apparent convergence tracing to a shared source rather than to independent roads dissolves and carries no warrant. The twelve-gate cascade: twelve directed failure screens, self-reference, frame-dependence, missing mechanism, and the rest, run in order, the first failure terminating the verdict with its mechanism named.

Box G1 | The lock kernel identity, proved

The lock is a determinant identity, not a metaphor. Let the three axes, after normalization, be unit vectors expressed in an orthonormal basis of the subspace they span and read as pure quaternions a, b, c , of norm one each. Hamilton's product of two pure quaternions is $p q = -(p \cdot q) + p \times q$, the real part the negative dot product and the imaginary part the cross product, checked on the units by $i j = k = i \times j$ and $i i = -1$. The lock scalar is the real part of the triple product, $\lambda = \text{Re}(a b c)$.

Expanding, $a b = -(a \cdot b) + a \times b$, so $\text{Re}(a b c) = -(a \cdot b) \text{Re}(c) + \text{Re}((a \times b) c)$. The first term vanishes since c is pure, and the second is $-(a \times b) \cdot c$ by the same product rule, giving $\lambda = -(a \times b) \cdot c = -\det[a b c]$, minus the signed volume of the parallelepiped the three axes span. Writing A for the matrix whose columns are a, b, c , the correlation matrix is $R = A^T A$ with entries the pairwise dot products, so $\det(R) = \det(A^T A) = \det(A)^2 = \lambda^2$.

The identity $\lambda^2 = \det(R)$ states that the squared signed volume equals the Gram determinant, with the quaternion triple product the machine that computes the volume. Executed at seed 20260622 over twenty thousand random triads at twenty-four contexts, the residual $|\lambda^2 - \det(R)|$ reached a maximum of 4.8×10^{-15} , and the observed determinants ranged over the interval from 0.273 to 0.9998, inside the theoretical bounds.

Four consequences fix the reading. The determinant is bounded, $\det(R)$ in the interval from zero to one for unit axes, by Hadamard's inequality above and positive-semidefiniteness below. The floor is the break: $\det(R)$ equals zero exactly when the three axes are linearly dependent, one axis lying in the plane of the other two. The magnitude is orientation-blind: reflecting any axis flips the sign of λ while $\det(R) = \lambda^2$ is unchanged, so the number certifies the dimensionality of the lock and never its truth-sign, which is read from the ordered axes and not from the scalar. And the magnitude is frame-invariant: rotating all three axes together is conjugation on the imaginary quaternions, under which the real part is preserved.

The axis count is three because the algebra forces it. The audit-composition law requires associativity, so iterated audits bracket the same way, and the absence of zero divisors, so nonzero warrants never compound to nothing. By Frobenius's theorem the only finite-dimensional associative division algebras over the reals are the reals, the complex numbers, and the quaternions, and three mutually orthogonal imaginary axes exist only in the quaternions. The next normed division algebra, the octonions, fails associativity, witnessed by the non-zero associator of e_1, e_2, e_4 , which equals $2e_7$, so it cannot carry the composition law and the construction stops at three.

This paper in particular. The three axes as instantiated are the formal constraint the framework imposes at a future coordinate, the present state propagated forward by identified dynamics, and the registration, which is the measurement and is empty until it is made. The decisive result is a negative one and it is the paper's Section 3: executed at seed 20260622, a manifest plane at $\sin^2(\theta) = 0.859132$ returns $\det(R) = 0.853265$ for a witness aligned to an independent generator and

0.849903 for a witness drawn from noise, a separation of 0.0034, so the lock scalar is constant on the orthogonal complement and cannot discriminate provenance; and over two hundred thousand random centred pairs at twenty-four contexts, 0.9999 of pure noise clears a threshold of one half, so an orthogonality floor admits everything. The load-bearing step is therefore not the lock but the negation test at the last stage of the cascade, which asks whether the structure forbids the denial, and which alone produced the reduction from seventy-seven candidates to five. The verdict is open on all five standing predictions, at structural grade, with the consequence of each stated and two apertures named; the method claim, that exposure is measurable by typing rather than by counting, is structural and not theorem-grade. No new mathematics is claimed. Every result used is classical and cited, the contribution is the arrangement and the typing, and the change in mathematical mass is zero.

The canonical statement of the method, its axioms, and its executable batteries lives in the reference cited below, continuously updated at the same location. This note is a pointer, not a substitute, and nothing in this appendix is a step in the paper's argument.

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Data and reproduction

The admission cascade, its declared inputs, the disposition ledger, and the two numerical exhibits are executable and were run at a fixed seed. The orthogonality distribution used two hundred thousand random centred vector pairs at N equal to twenty-four; the source-blindness exhibit used one sourced and six unsourced third axes on a common manifest plane. All are reproducible from the published code and ledger.

Competing interests and author provenance

The author declares no competing financial interests.

Three non-financial dependences are declared because each bears on how the results should be weighted. The framework evaluated is the author's own, and the audit that produced the reduction was performed by the author rather than by an independent party. The cascade's steps were formalised after the candidate register existed and were not pre-registered against it, so neither the entry conditions nor the decisive negation test were blind to the material they filtered; what is pre-registered is the set of kill conditions, fixed at publication and not revised against incoming data. And the corpus from which this paper is drawn is assembled with the assistance of large language models operating under the author's direction and seeded inputs, with the author responsible for every claim, every typing decision, and every figure; all numerical exhibits were executed rather than recalled, at the declared seed, and are reproducible from the published scripts.

The adversarial self-audit that shaped this version was run under the same single-substrate condition and is disclosed as such. Its ledger, carrying every finding raised and its disposition, is published alongside so that a reader may verify what was found, what was repaired, what was scoped, and what remains open.

Author contributions

M.F.I. developed the method, performed the audit, executed the numerical exhibits, and wrote the manuscript.